

## Homework12

### Question 18.1

This is my approach to a comprehensive, deep-dive mathematical and structural blueprint for the Power Company Analytics Case. The framework laid out below tries to bridge the gap between statistical prediction, financial cost-benefit analysis and operational research to create an end-to-end prescriptive model.

The reason I state it as prescriptive model is 1) We are suggesting/proposing an optimal method to perform shut offs 2) We are also part predicting on which units to be shutoff in each month within a capacity constraint.

### **Phase 1: The Predictive Component (Logistic Regression)**

Before we can route a truck and determine the number of shut offs per month, we must predict the likelihood that a delinquent customer will self-correct (pay their bill without intervention). *Given that “all customers who miss their payments are not guaranteed to be a delinquent customer”, use “classification(predictive) either logistic or SVM” to “predict the probability that a customer will pay their bill without a shut off”.* This identifies “self-correcting” accounts that do not require an expensive field visit.

1. **The Goal:** Calculate  $P_i$ , which is the probability that customer  $i$  will pay their bill organically within the next grace period.
2. **The Logistic Regression Equation:** We model the probability using the sigmoid function.

$$P_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki})}}$$

3. **The Feature Vector ( $X$ ):**

To build the linear combination inside the exponent, we define specific variables:

- $X_1$  : **Account Tenure** (Months since the customer opened the account).
- $X_2$  : **Historical Delinquency Rate** (Percentage of past bills paid late).
- $X_3$  : **Cure Rate** (Number of times they are late but paid before a shutoff).
- $X_4$  : **Debt Ratio** (Current amount owed divided by their average historical monthly bill).
- $X_5$  : **Service Plan Type** (Current type of service plan).

The coefficients ( $\beta$ ) are “trained” on historical power company data. A high  $P_i$  means the customer is highly likely to pay without us doing anything.

## Phase 2: The Cost of Error Component (Financial Modeling)

We cannot simply shut off everyone with a low  $P_i$ . We must convert that probability into a **Net Expected Value (NEV)** to account for the costs of making a mistake. *Given that “the power company does not infinite resources and there is cost incurred”, use “a ranking model to order customers” to “calculate the expected financial value of executing a shutoff versus doing nothing”.*

### 1. Defining the Financial Parameters:

- $V_i$  : Total debt owed by customer  $i$ .
- $C_s$  : Logistical cost of a shutoff (truck fuel, technician hourly wage).
- $C_p$  : The “Penalty” cost of a False Positive (lost future revenue, customer service calls, or regulatory fines for shutting off a customer who was actively paying).

### 2. The Confusion Matrix Outcomes:

- **True Positive (TP):** We predict they won’t pay; we shut them off; we force the recovery of  $V_i$ . The net gain is  $V_i - C_s$ .
- **False Positive (FP):** We predict they won’t pay, we shut them off, but they were going to pay. We wasted the truck roll and angered the customer. The net loss is  $(-C_s - C_p)$ .
- **True Negative (TN):** We predict they will pay; we do nothing, they pay. Net gain is  $V_i$ .
- **False Negative (FN):** We predict they will pay; we do nothing, they default. We lose  $V_i$ .

### 3. The Confusion Matrix Outcomes:

For every customer as stated above, we calculate the expected financial value of executing a shutoff versus doing nothing.

$$\text{Expected Value of Shutoff} = (1 - P_i)(V_i - C_s) + P_i(-C_s - C_p)$$

$$\text{Expected Value of Doing Nothing} = P_i(V_i) + (1 - P_i)(0)$$

We only add customer  $i$  to our “Candidate Shutoff List” if the Expected Value of Shutoff is greater than the Expected Value of Doing Nothing. This is our mathematical threshold for action. Let  $E_i$  be the isolated recovery value of a justified shutoff for customer  $i$ . This is simplified heuristic which represent avoidable loss where  $(1 - P_i)V_i$  represents the amount of debt we expect to lose forever if we don’t act (it is the risk associate with customer  $i$ ) and  $C_s$  is the cost to eliminate that risk:

$$E_i = (1 - P_i)V_i - C_s$$

So,  $E_i$  tells the optimization model which is coming next “How much debt can I save by spending  $C_s$  on this specific customer?”

### Phase 3: The Prescriptive Component (Optimization)

This part is like or can draw parallel with the vehicle routing problem (VRP) or Travelling Salesman Problem (TSP) and we need to determine the most efficient path for technicians, maximizing the number of shutoffs performed within the travel time constraints. *Given “a candidate list of  $N$  customers where a shutoff makes financial sense, each with an expected value of  $E_i$ ” use “linear programming optimization model with variables, constraints and objective function” to “identify the optimal routing under the limited resources (number of technicians, finite hours in a day) to maximize the total expected recovery debt”.* This is modeled as a prize-collecting vehicle routing problem:

#### 1. Sets and Parameters:

- $N$  : Set of all candidate customer locations, plus the depot (node 0).
- $K$  : Set of all available field technicians.
- $T_{max}$  : Maximum shift time (e.g., 8 hours).
- $t_{ij}$  : Travel time from location  $i$  to location  $j$ .
- $s_i$  : Service time required to physically shut off the meter at location  $i$ .

#### 2. Decision Variables:

- $y_i \in \{0, 1\}$  : Equals 1 if we decide to shut off customer  $i$ , 0 otherwise.
- $x_{ijk} \in \{0, 1\}$  : Equals 1 if technician  $k$  travels directly from node  $i$  to node  $j$ .

#### 3. The Objective Function:

Maximize the total expected recovered debt:

$$\text{Maximize } Z = \sum_{i \in N} E_i \cdot y_i$$

#### 4. The Operational Constraints:

- **Time/Capacity Constraint:** A technician cannot work longer than their shift. Travel time plus service time must be less than  $T_{max}$  :

$$\forall k \in K : \sum_{i \in N} \sum_{j \in N} (t_{ij} \cdot x_{ijk}) + \sum_{i \in N} (s_i \cdot y_i) \leq T_{max}$$

- **Routing/Flow Constraint:** If a technician arrives at a location, they must leave that location:

$$\forall j \in N, \forall k \in K : \sum_{i \in N} x_{ijk} = \sum_{m \in N} x_{jmk} = y_j$$

- **Single Visit Constraint:** A location can be visited by a maximum of one technician:

$$\forall i \in N : \sum_{k \in K} \sum_{j \in N} x_{ijk} = y_j \leq 1$$

- **Subtour Elimination Constraint:** The solver finds that the shortest distance is to visit 5 houses in a perfect circle in one neighborhood and the other 5 in a circle in another neighborhood. Mathematically, every house has one entrance, and one exist, so the flow equations above will be satisfied. But physically, the technician is stuck in one loop and can never reach the other and neither loop is connected to the power company headquarters (the depot). Hence the common way to fix this is to assign a “sequence number” or “rank” to each stop. If a technician travels from house  $i$  to house  $j$ , then house  $j$  must be the “next” stop in the sequence and therefore the rank of  $j$  must be greater than rank of  $i$ . Because the numbers must always increase (the rank number), we can never loop back to house we’ve already visited unless we go through the depot (where the sequence resets). This effectively “breaks” any potential subtours.

## Phase 4: System Integration (The “Finer Details”)

The overall solution answered the following - *Given “customer payment history, debt amounts, and field technician capacity” use “a combination of a Logistic Regression classifier and a linear programming optimization model” to “identify the optimal set of monthly shutoffs that maximizes debt recovery while minimizing travel time and operational costs”.* From the above methodology of implementation, the following questions have been integrated. 1) The model must prioritize accounts where the “propensity to pay” is low but the “recovery value” is high. 2) Given that travel consumes worker capacity, the model must “cluster” shutoffs geographically. A high-priority shutoff in a remote area might be less valuable than four medium priority shutoffs in a tight city block. 3) The predictive model must specifically flag customers who “just forgot” or have a high historical probability of paying within 10 days of a late notice. These accounts are “Capacity Free” recoveries – meaning you get the money without spending worker time.

To put this into production, the pipeline runs sequentially every morning:

1. **Batch Processing:** Run yesterday’s billing data through the Logistic Regression model to update all  $P_i$  values.
2. **Threshold filtering:** Run the Cost of Error equations. Drop all customers from the list where the expected value of doing nothing exceeds the value of a shutoff.
3. **Geospatial Matrix Generation:** Use an API (like Google Maps Distance Matrix) to calculate the real-world driving times ( $t_{ij}$ ) between all surviving candidates.

4. **Solver Execution:** Feed the  $E_i$  values and  $t_{ij}$  matrix into a Mixed-Integer Linear Programming (MILP) solver (like PuLP in Python) to maximize  $Z$ .
5. **Output:** The solver outputs the exact  $x_{ijk}$  paths. These are pushed to the technicians tablets as their daily route.